(Membrane)-Cortex Mechanochemical FEM

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1 Elastic 2D energy for the membrane

1.1 Preliminary geometrical parametrization

We consider a material defined by the position vector $\boldsymbol{X}_0$ in its natural state (also called stress-free or reference state) and $\boldsymbol{x}\big(\boldsymbol{X}_0, t\big)$ in the current state. The deformation from the reference state is measured by a deformation gradient tensor defined by $d\boldsymbol{x} \equiv \boldsymbol{F} \cdot d\boldsymbol{X}_0$, that is

$$\underline{\underline{F}}(\boldsymbol{X}_0, t) = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}_0} \tag{1}$$

The scalar product after deformation between two infinitesimal material vectors is measured by the right Cauchy-Green deformation tensor defined by $d\boldsymbol{x}' \cdot d\boldsymbol{x} \equiv d\boldsymbol{X}_0' \cdot \underline{\underline{C}} \cdot d\boldsymbol{X}_0$, which yields

$$\underline{\underline{C}} = \underline{\underline{F}}^T \cdot \underline{\underline{F}} \tag{2}$$

In the reference state the membrane is described by the 3-dimensional position vector $\boldsymbol{X}_0 = R_0\, \boldsymbol{e}_r(\theta, \varphi)$ where R_0 is the stress-free radius. The spherical membrane is first inflated isotropically to an intermediate state $\boldsymbol{X} = R\, \boldsymbol{e}_r = \xi\, \boldsymbol{X_0}$, where $\xi \equiv R/R_0$ is the isotropic biaxial stretch for this intermediate deformation, putting it under basal tension. In the following, we call ξ the **membrane prestretch**. The current membrane shape is ultimately described by the position vector

$$\boldsymbol{x}(\boldsymbol{X}_0, t) = \boldsymbol{X} + \boldsymbol{u}(\boldsymbol{X}_0, t) = \xi \boldsymbol{X}_0 + \boldsymbol{u}(\boldsymbol{X}_0, t) \tag{3}$$

where $\boldsymbol{u}(\theta, \varphi, t) = \big(u_r, u_\theta, u_\varphi\big)$ is a 3-dimensional displacement vector from the intermediate state, which depends on the polar and azimuthal angles $\big(\theta, \varphi\big)$.

The deformation gradient $\underline{\underline{F}}$ is a map from the plane tangent to the reference configuration, to the 3D Eucliean space. These two spaces are endowed with the bases $(\boldsymbol{e}_\theta, \boldsymbol{e}_\varphi)$ and $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_\varphi)$, respectively. Then, $\underline{\underline{F}}$ can be represented as a 3×2 matrix,

$$\underline{\underline{F}} = \underline{\underline{\boldsymbol{\nabla}}}_{\boldsymbol{X}_0}(\boldsymbol{x}) = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}_0} = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}} \cdot \frac{\partial \boldsymbol{X}}{\partial \boldsymbol{X}_0} = \xi\, \left(\underline{\underline{\mathbb{1}}}_{3\times 2} + \underline{\underline{\boldsymbol{\nabla}}}_{\boldsymbol{X}}(\boldsymbol{u})\right)\,, \tag{4}$$

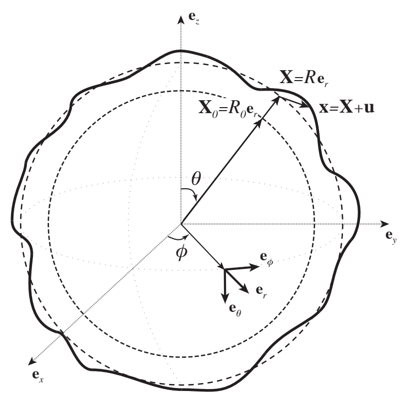

Figure 1: **Figure e**: The fluctuating elastic membrane (bold line) is described by a position vector $\boldsymbol{x} = \boldsymbol{X} + \boldsymbol{u}$, where the displacement vector $\boldsymbol{u}$ from intermediate state is projected on the vector basis $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_\varphi)$. In reference state, the membrane lies on the sphere $\boldsymbol{X}_0 = R_0\boldsymbol{e}_r$, and in the intermediate prestretched state, it is described by the spherical position vector $\boldsymbol{X} = R\boldsymbol{e}_r = \xi R_0 \boldsymbol{e}_r$.

where we have defined the matrix $\underline{\underline{\mathbb{1}}}_{3\times 2} \equiv \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ representing the embedding of the tangent plane into the 3-dimensional Euclidean space. The gradient $\underline{\underline{\boldsymbol{\nabla}}}_{\boldsymbol{X}}(\boldsymbol{u})$ is defined with respect to the intermediate state $\boldsymbol{X}$ and we write it $\underline{\underline{\boldsymbol{\nabla} \boldsymbol{u}}}$ in the following to simplify the notations.

The right Cauchy-Green deformation tensor is given by

$$\begin{aligned} \underline{\underline{C}} &= \underline{\underline{F}}^T \cdot \underline{\underline{F}} \\ &= \xi^2 \left[\underline{\underline{\mathbb{1}}}_{2\times 2} + \underline{\underline{\boldsymbol{\nabla}_{\mathrm{s}} \boldsymbol{u}}} + \underline{\underline{\boldsymbol{\nabla}_{\mathrm{s}} \boldsymbol{u}}}^T + \underline{\underline{\boldsymbol{\nabla} \boldsymbol{u}}}^T \cdot \underline{\underline{\boldsymbol{\nabla} \boldsymbol{u}}} \right] \\ &\equiv \xi^2 \left[\underline{\underline{\mathbb{1}}}_{2\times 2} + 2\underline{\underline{e}} \right] \end{aligned} \tag{5}$$

where we have introduced the incremental finite strain tensor

$$\underline{\underline{e}} = \frac{1}{2} \left(\underline{\underline{\boldsymbol{\nabla}_{\mathrm{s}} u}} + \underline{\underline{\boldsymbol{\nabla}_{\mathrm{s}} u}}^T \right) + \frac{1}{2} \underline{\underline{\boldsymbol{\nabla} u}}^T \cdot \underline{\underline{\boldsymbol{\nabla} u}} \tag{6}$$

The surface gradient $\boldsymbol{\nabla}_{\mathrm{s}} = \underline{\underline{\mathbb{1}}}^T_{3\times 2} \cdot \boldsymbol{\nabla} = \boldsymbol{\nabla}_{2\times 2} = \frac{\boldsymbol{e}_\theta}{r} \frac{\partial}{\partial_\theta} + \frac{\boldsymbol{e}_\varphi}{r \sin\theta} \frac{\partial}{\partial_\varphi}$ is the projection of the 3-dimensional gradient onto the plane tangent to the membrane in the intermediate configuration.

The displacement vector $\boldsymbol{u}$ and strain tensor $\underline{\underline{e}}$ are said *incremental*, because they are defined with respect to the prestressed intermediate state.

1.2 Incremental linear elastic energy density for an isotropic 2D material

We show below that for any isotropically prestressed 2-dimensional elastic material, we always obtain the same energy density form to second order in the incremental strain tensor $\underline{\underline{e}}$: $\mathcal{E} = \mathcal{E}_\mathrm{i} + S\,\mathrm{tr}\left(\underline{\underline{e}}\right) + \frac{\Lambda}{2}\,\mathrm{tr}\left(\underline{\underline{e}}\right)^2 + M\,\mathrm{tr}\left(\underline{\underline{e}}^2\right)$ (Hookean law with prestress [1]).

For 2-dimensional isotropic finite elasticity, the elastic energy density $\mathcal{E}$ is a function of the 2 invariants only

$$I_1 = \mathrm{tr}\left(\underline{\underline{C}}\right) \qquad I_2 = \det\left(\underline{\underline{C}}\right) \tag{7}$$

These two invariants can be expressed as a function of the principal stretch ratios λ_1 and λ_2 as

$$I_1 = \lambda_1^2 + \lambda_2^2 \qquad I_2 = \lambda_1^2\lambda_2^2$$

We express the invariants as function of the prestretch ratio ξ and the incremental strain tensor $\underline{\underline{e}}$

$$I_1 = \mathrm{tr}\left(\underline{\underline{C}}\right) = 2\xi^2\left[1 + \mathrm{tr}\,\underline{\underline{e}}\right] \tag{8a}$$

$$\begin{aligned} I_2 = \det\left(\underline{\underline{C}}\right) &= \det\left[\xi^2\left(1 + 2\,\mathrm{tr}\,\underline{\underline{e}}\right)\right] = \xi^4\left(1 + \mathrm{tr}\left(2\underline{\underline{e}}\right) + \det\left(2\underline{\underline{e}}\right)\right) \\ &= \xi^4\left[1 + 2\,\mathrm{tr}\left(\underline{\underline{e}}\right) + 4\,\det\left(\underline{\underline{e}}\right)\right] \end{aligned} \tag{8b}$$

In 2 dimensions, $\det\left(\underline{\underline{e}}\right) = e_{11}e_{22} - e_{12}^2 = \frac{1}{2}\left[\mathrm{tr}^2\left(\underline{\underline{e}}\right) - \mathrm{tr}\left(\underline{\underline{e}}^2\right)\right]$ and we deduce

$$I_2 = \xi^4\left[1 + 2\mathrm{tr}\left(\underline{\underline{e}}\right) + 2\mathrm{tr}^2\left(\underline{\underline{e}}\right) - 2\mathrm{tr}\left(\underline{\underline{e}}^2\right)\right] \tag{9}$$

We can develop formally the energy $\mathcal{E}\left(I_1\left(\underline{\underline{e}}\right), I_2\left(\underline{\underline{e}}\right)\right)$ to 2nd order in $\underline{\underline{e}}$

$$\begin{aligned} \mathcal{E} = \mathcal{E}_\mathrm{i} &+ \mathcal{E}_{,1}\left(2\xi^2\,\mathrm{tr}\left(\underline{\underline{e}}\right)\right) + \mathcal{E}_{,2}\,2\xi^4\left(\mathrm{tr}\left(\underline{\underline{e}}\right) + \mathrm{tr}^2\left(\underline{\underline{e}}\right) - \mathrm{tr}\left(\underline{\underline{e}}^2\right)\right) \\ &+ \frac{1}{2}\mathcal{E}_{,11}\left(2\xi^2\mathrm{tr}\left(\underline{\underline{e}}\right)\right)^2 + \mathcal{E}_{,12}\left(2\xi^2\mathrm{tr}\left(\underline{\underline{e}}\right)\right)\left(2\xi^4\mathrm{tr}\left(\underline{\underline{e}}\right)\right) + \frac{1}{2}\mathcal{E}_{,22}\left(2a^4\mathrm{tr}\left(\underline{\underline{e}}\right)\right)^2 + \mathcal{O}\left(|\underline{\underline{e}}|^2\right) \end{aligned} \tag{10}$$

I don't understand these definitions, but no need to, see (23).

where we note $\mathcal{E}_\mathrm{i} \equiv E\left(2\xi^2, \xi^4\right)$, $\mathcal{E}_{,i} \equiv \partial_{I_i}\mathcal{E}|_{2\xi^2,\xi^4}$ and $\mathcal{E}_{,ij} \equiv \partial_{I_i}\partial_{I_j}\mathcal{E}|_{2\xi^2,\xi^4}$.

By grouping terms, we find a **Hookean law with prestress**:

$$\mathcal{E} = \mathcal{E}_\mathrm{i} + S\,\mathrm{tr}\,\underline{\underline{e}} + \frac{1}{2}\Lambda\,\mathrm{tr}^2\left(\underline{\underline{e}}\right) + M\,\mathrm{tr}\left(\underline{\underline{e}}^2\right) \tag{11}$$

where

$$S = 2\xi^2\,\mathcal{E}_{,1} + 2\xi^4\,\mathcal{E}_{,2} \tag{12a}$$

$$\Lambda = 4\xi^4\,\mathcal{E}_{,2} + 4\xi^4\,\mathcal{E}_{,11} + 8\xi^6\,\mathcal{E}_{,12} + 4\xi^8\,\mathcal{E}_{,22} \tag{12b}$$

$$M = -2\xi^4\,\mathcal{E}_{,2} \tag{12c}$$

1.3 Homogeneization of an hexagonal network of spring

Kantor noted that, from the point of view of elasticity, a sixfold symmetry is equivalent to isotropic elasticity [2]. To make the link with the 1D model, we therefore homogeneize below a 2D hexagonal network of springs.

Preliminaries We start with the homogenization procedure, that is done in the tangential plane of the membrane, so in effectively in 2-dimensions.

We note α the angle that a spring makes with $\boldsymbol{e}_\theta$ in reference configuration in the 2-dimensional network plane and $\boldsymbol{t}(\alpha) = \big(\cos\alpha,\, \sin\alpha\big)$ it unitary tangent. Noting $\langle\cdot\rangle_\alpha$ the average over the six directions α of the hexagonal network, or equivalently over all angles α, one can calculate the following second-order and fourth-order tensors [3]

$$\underline{\underline{I}}_2 = \big\langle \boldsymbol{t}\otimes\boldsymbol{t}\big\rangle_\alpha = \frac{1}{2}\,\underline{\mathbb{1}}_{2\times 2} \tag{13a}$$

$$\underline{\underline{\underline{\underline{I}}}}_4 = \big\langle \boldsymbol{t}\otimes\boldsymbol{t}\otimes\boldsymbol{t}\otimes\boldsymbol{t}\big\rangle_\alpha = 2\tilde{\mu}\,\underline{\underline{\underline{\mathbb{1}}}}^4_S + \tilde{\lambda}\,\underline{\mathbb{1}}_{2\times 2}\otimes\underline{\mathbb{1}}_{2\times 2} \tag{13b}$$

where the tensors introduced above are defined as follows in the membrane plane (see [3])

$$\big(\underline{\mathbb{1}}_{2\times 2}\big)_{ij} = \delta_{ij}$$

$$\big(\underline{\mathbb{1}}_{2\times 2}\otimes\underline{\mathbb{1}}_{2\times 2}\big)_{ijkl} = \delta_{ij}\,\delta_{kl}$$

$$\Big(\underline{\underline{\underline{\mathbb{1}}}}^4_S\Big)_{ijkl} = \frac{1}{2}\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big)$$

and we can calculate the coefficients $\tilde{\lambda} = \tilde{\mu} = \frac{1}{8}$.

We deduce the following useful expressions for the following (for details see [3])

$$\underline{\underline{e}} : \underline{\underline{I}}_2 = \frac{1}{2}\,\mathrm{tr}\,\big(\underline{\underline{e}}\big) \tag{15a}$$

$$\underline{\underline{e}} : \underline{\underline{\underline{\underline{I}}}}_4 : \underline{\underline{e}} = 2\tilde{\mu}\,\big(\underline{\underline{e}} : \underline{\underline{e}}\big) + \tilde{\lambda}\,\mathrm{tr}^{\,2}\big(\underline{\underline{e}}\big) \tag{15b}$$

Homogenization

We consider an isotropic network of filaments having identical natural lengths

ℓ_0. They undergo the following quasi-isotropic deformation from their natural state that we described previously:

$$\underset{\text{reference 2D}}{\boldsymbol{X}_0} \xrightarrow{\times\,\xi} \underset{\text{intermediate 2D}}{\boldsymbol{X}} \xrightarrow{+\,\boldsymbol{u}(\boldsymbol{X})} \underset{\text{final 3D}}{\boldsymbol{x}}$$

As we have before, this deformation is described by the deformation gradient tensor $\underline{\underline{F}} = \xi\Big(\mathbb{1}_{3\times 2} + \underline{\underline{\nabla}}_{\boldsymbol{X}}\boldsymbol{u}\Big)$ (4) and the resulting Cauchy-Green tensor is $\underline{\underline{C}} = \xi^2\Big(\mathbb{1}_{2\times 2} + 2\underline{\underline{e}}\Big)$ (5).

We introduce ρ_0 the surface density of filaments in the network and $e\big(\lambda\big)$ the elastic energy of one filament of stretch ratio $\lambda = \ell/\ell_0$. On length-scales larger than the mesh-size $\langle\ell\rangle$, the isotropic network discrete elastic energy can be homogenized into the following continuous form

$$E = \int \big\langle e(\lambda)\big\rangle_\alpha\, \rho_0\, dA_0 \tag{16}$$

Introducing $\boldsymbol{t}_0$ the filament unit tangent in reference configuration, the filament stretch ratio λ is given by

$$\lambda = \Big|\underline{\underline{F}}\cdot\boldsymbol{t}_0\Big| = \sqrt{\boldsymbol{t}_0\cdot\underline{\underline{C}}\cdot\boldsymbol{t}_0} = \xi\big(1+2e_{tt}\big)^{1/2} \tag{17}$$

where $e_{tt} \equiv \boldsymbol{t_0}\cdot\underline{\underline{e}}\cdot\boldsymbol{t}_0$.

We expand the homogenized filament elastic energy up to second order in e_{tt}

$$\big\langle e(\lambda)\big\rangle_\alpha \sim \Big\langle e\big(\xi\sqrt{1+2e_{tt}}\big)\Big\rangle_\alpha = e(\xi) + \xi\, e'(\xi)\, \langle e_{tt}\rangle_\alpha + \frac{1}{2}\xi^2\left(e''(\xi) - \frac{1}{\xi}e'(\xi)\right)\langle e_{tt}^2\rangle_\alpha \tag{18}$$

With the help of the preliminary calculations (15), we get

$$\begin{aligned}\big\langle e_{tt}\big\rangle_\alpha &= \big\langle \boldsymbol{t}_0\cdot\underline{\underline{e}}\cdot\boldsymbol{t}_0\big\rangle_\alpha = \underline{\underline{e}} : \big\langle \boldsymbol{t}_0\otimes\boldsymbol{t}_0\big\rangle_\alpha \\ &= \underline{\underline{e}} : \frac{1}{2}\mathbb{1}_{2\times 2} = \frac{1}{2}\mathrm{tr}\,\big(\underline{\underline{e}}\big) \end{aligned} \tag{19a}$$

$$\begin{aligned}\big\langle e_{tt}^2\big\rangle_\alpha &= \big\langle \boldsymbol{t}_0\otimes\boldsymbol{t}_0\otimes\boldsymbol{t}_0\otimes\boldsymbol{t}_0\big\rangle :: \big(\underline{\underline{e}}\otimes\underline{\underline{e}}\big) = \underline{\underline{I}}_4 :: \big(\underline{\underline{e}}\otimes\underline{\underline{e}}\big) \\ &= \frac{1}{8}\Big(2\,\underline{\underline{e}} : \underline{\underline{e}} + \mathrm{tr}^{\,2}\big(\underline{\underline{e}}\big)\Big)\end{aligned} \tag{19b}$$

With these expressions, we obtain the following homogenized energy

$$E = \int \left\{ e(\xi) + \left(\frac{1}{2}\xi\, e'(\xi)\right)\mathrm{tr}\,\big(\underline{\underline{e}}\big) + \frac{1}{2}\left[\frac{1}{8}\xi^2\left(e''(\xi) - \frac{e'(\xi)}{\xi}\right)\right]\Big(2\mathrm{tr}\,\big(\underline{\underline{e}}^2\big) + \mathrm{tr}^{\,2}\big(\underline{\underline{e}}\big)\Big)\right\}\rho_0\, dA_0$$

that we can rewrite in the classical form of a Hookean energy with prestress [1]

$$E = \int \left[\mathcal{E}_i + S\,\mathrm{tr}\,\big(\underline{\underline{e}}\big) + \left(\frac{\Lambda}{2}\mathrm{tr}^{\,2}\big(\underline{\underline{e}}\big) + M\,\mathrm{tr}\,\big(\underline{\underline{e}}^2\big)\right)\right] dA \tag{20}$$

where we have introduced the area in the intermediate configuration $dA = \xi^2\, dA_0$ and the following effective incremental elastic moduli

$$\mathcal{E}_\mathrm{i}(\xi) = \rho_0\, e(\xi)/\xi^2 \tag{21a}$$

$$S(\xi) = \frac{1}{2}\rho_0\, e'(\xi)/\xi \tag{21b}$$

$$\Lambda(\xi) = M(\xi) = \frac{1}{8}\rho_0 \left(e''(\xi) - \frac{e'(\xi)}{\xi} \right) \tag{21c}$$

Our derivation remains very general, and does not depend on the homogenization or on the filament elasticity as we showed in section 1.2. Only the energy density $\mathcal{E}_\mathrm{i}$ and of the incremental elastic moduli S, Λ and M will vary as function of the prestretch ξ and mechanical parameters of the network in reference configuration.

Hexagonal network of linear springs

In the following, we make the simplifying hypothesis that the membrane may is a generalization of our 1D model as a perfect hexagonal network of linear springs.

For a regular network, the filament density is given by $\rho_0 = \frac{g}{\ell_0^2}$, where the metric g is a constant which depends on the meshwork topology: $g = 2\sqrt{3}$ for a hexagonal network, $g = 2$ for a square network.

In the particular case of an hexagonal network of linear springs, the filament energy reads $e(\lambda) = \frac{k}{2}\ell_0^2\,(\lambda - 1)^2$ and we deduce immediately

$$e(\xi) = \frac{k}{2}\ell_0^2(\xi - 1)^2 \qquad e'(\xi) = k\ell_0^2(\xi - 1) \qquad e''(\xi) = k\ell_0^2 \tag{22}$$

The intermediate energy and incremental elastic moduli in (21) can hence be expressed explicitly as a function of the spring stiffness k and network prestress ξ

$$\mathcal{E}_\mathrm{i}(\xi) = k\sqrt{3}\big(1 - 1/\xi\big)^2 \tag{23a}$$

$$S(\xi) = k\sqrt{3}\,\big(1 - 1/\xi\big) \tag{23b}$$

$$\Lambda(\xi) = M(\xi) = \frac{\sqrt{3}}{4}k/\xi \tag{23c}$$

Note that the above calculation extends to an isotropically prestressed elastic network the equivalence derived by Kantor between an hexagonal network of springs and the Hookean energy, for which the Lamé coefficients are shown to verify [2]

$$\lambda_0 = \mu_0 = \frac{\sqrt{3}}{4}k \tag{24}$$

The shear modulus in reference state $\mu_0 = \frac{\sqrt{3}}{4}k$ can alternatively be used as single parameter to characterize the membrane mechanics.

1.4 Energy as function of (incremental) displacement

In spherical geometry, the incremental deformation gradient $\underline{\underline{\nabla u}}$ is given by

$$\underline{\underline{\nabla u}} = \frac{1}{R}\begin{pmatrix} u_{r\theta} & u_{r\varphi} \\ u_{\theta\theta} & u_{\theta\varphi} \\ u_{\varphi\theta} & u_{\varphi\varphi} \end{pmatrix} = \frac{1}{R}\begin{pmatrix} \frac{\partial u_r}{\partial\theta} - u_\theta & \frac{1}{\sin\theta}\frac{\partial u_r}{\partial\varphi} - u_\varphi \\ u_r + \frac{\partial u_\theta}{\partial\theta} & \frac{1}{\sin\theta}\frac{\partial u_\theta}{\partial\varphi} - \cot\theta\, u_\varphi \\ \frac{\partial u_\varphi}{\partial\theta} & u_r + \frac{1}{\sin\theta}\frac{\partial u_\varphi}{\partial\varphi} + \cot\theta\, u_\theta \end{pmatrix} \tag{25}$$

We have therefore

$$\frac{\underline{\underline{\nabla_s u}}^T + \underline{\underline{\nabla_s u}}}{2} = \frac{1}{R}\begin{pmatrix} u_{\theta\theta} & \frac{u_{\theta\varphi}+u_{\varphi\theta}}{2} \\ \frac{u_{\theta\varphi}+u_{\varphi\theta}}{2} & u_{\varphi\varphi} \end{pmatrix} \tag{26}$$

and $$\frac{1}{2}\underline{\underline{\nabla u}}^T \cdot \underline{\underline{\nabla u}} = \frac{1}{2R^2}\begin{pmatrix} u_{r\theta}^2+u_{\theta\theta}^2+u_{\varphi\theta}^2 & u_{r\theta}u_{r\varphi}+u_{\theta\theta}u_{\theta\varphi}+u_{\varphi\varphi}u_{\varphi\theta} \\ u_{r\theta}u_{r\varphi}+u_{\theta\theta}u_{\theta\varphi}+u_{\varphi\varphi}u_{\varphi\theta} & u_{r\varphi}^2+u_{\theta\varphi}^2+u_{\varphi\varphi}^2 \end{pmatrix} \tag{27}$$

We deduce, to second order in $\underline{\underline{\nabla u}}$

$$\operatorname{tr}\left(\underline{\underline{e}}\right) \sim \frac{1}{R}\left[\left(u_{\theta\theta}+u_{\varphi\varphi}\right) + \frac{1}{2R}\left(u_{\theta\theta}^2+u_{\varphi\varphi}^2+u_{r\theta}^2+u_{r\varphi}^2+u_{\theta\varphi}^2+u_{\varphi\theta}^2\right)\right] \tag{28a}$$

$$\operatorname{tr}^2\left(\underline{\underline{e}}\right) \sim \frac{1}{R^2}\left[\left(u_{\theta\theta}+u_{\varphi\varphi}\right)^2\right] \tag{28b}$$

$$\operatorname{tr}\left(\underline{\underline{e}}^2\right) \sim \frac{1}{R^2}\left[u_{\theta\theta}^2 + u_{\varphi\varphi}^2 + \frac{\left(u_{\theta\varphi}+u_{\varphi\theta}\right)^2}{2}\right] \tag{28c}$$

which gives the expression for the energy density $\mathcal{E}(\boldsymbol{u}, \tilde{\mu})$ as function of the membrane displacement $\boldsymbol{u}(\theta, \varphi, t)$.

Axisymmetric geometry Axisymmetric is no azimuthal + no normal = tangent

In axisymmetric geometry, supposing furthermore no normal nor azimuthal displacements $u_r = u_\varphi = 0$, this simplifies to

$$\underline{\underline{\nabla u}} = \frac{1}{R}\begin{pmatrix} u_{r\theta} & u_{r\varphi} \\ u_{\theta\theta} & u_{\theta\varphi} \\ u_{\varphi\theta} & u_{\varphi\varphi} \end{pmatrix} = \frac{1}{R}\begin{pmatrix} -u_\theta & 0 \\ \frac{\partial u_\theta}{\partial\theta} & 0 \\ 0 & u_r + \cot\theta\, u_\theta \end{pmatrix} \tag{29}$$

We have therefore

$$\frac{\underline{\underline{\nabla_s u}}^T + \underline{\underline{\nabla_s u}}}{2} = \frac{1}{R}\begin{pmatrix} u_{\theta\theta} & 0 \\ 0 & u_{\varphi\varphi} \end{pmatrix} \tag{30}$$

and $$\frac{1}{2}\underline{\underline{\nabla u}}^T \cdot \underline{\underline{\nabla u}} = \frac{1}{2R^2}\begin{pmatrix} u_{r\theta}^2+u_{\theta\theta}^2 & 0 \\ 0 & u_{\varphi\varphi}^2 \end{pmatrix} \tag{31}$$

We deduce, to second order in $\underline{\underline{\nabla u}}$

$$\underline{\underline{e}} = \frac{1}{R}\begin{pmatrix} u_{\theta\theta} & 0 \\ 0 & u_{\varphi\varphi} \end{pmatrix} + \frac{1}{2R^2}\begin{pmatrix} u_{r\theta}^2 + u_{\theta\theta}^2 & 0 \\ 0 & u_{\varphi\varphi}^2 \end{pmatrix} \tag{32a}$$

$$\operatorname{tr}\left(\underline{\underline{e}}\right) = \frac{1}{R}\left[(u_{\theta\theta} + u_{\varphi\varphi}) + \frac{1}{2R}\left(u_{\theta\theta}^2 + u_{\varphi\varphi}^2 + u_{r\theta}^2\right)\right] \tag{32b}$$

$$\operatorname{tr}^2\left(\underline{\underline{e}}\right) = \frac{1}{R^2}\left[(u_{\theta\theta} + u_{\varphi\varphi})^2\right] \tag{32c}$$

$$\operatorname{tr}\left(\underline{\underline{e}}^2\right) = \frac{1}{R^2}\left[u_{\theta\theta}^2 + u_{\varphi\varphi}^2\right] \tag{32d}$$

1.5 Summary

In the next, we will consider for the membrane the following generic linearly elastic energy:

$$\mathcal{E} = \int \left[\mathcal{E}_i + S \operatorname{tr}\left(\underline{\underline{e}}\right) + \left(\frac{\Lambda}{2}\operatorname{tr}^2\left(\underline{\underline{e}}\right) + M \operatorname{tr}\left(\underline{\underline{e}}^2\right)\right)\right] dA \tag{33}$$

where $\underline{\underline{e}} = \frac{1}{2}\left(\underline{\underline{\boldsymbol{\nabla}_{\mathrm{s}} u}} + \underline{\underline{\boldsymbol{\nabla}_{\mathrm{s}} u}}^T\right) + \frac{1}{2}\underline{\underline{\boldsymbol{\nabla} u}}^T \cdot \underline{\underline{\boldsymbol{\nabla} u}}$ is a geometrically non-linear strain.

The 2-dimensional membrane stress tensor can be simply derived as

$$\underline{\underline{N}}^{\mathrm{mbr}} = \frac{\partial \mathcal{E}}{\partial \underline{\underline{e}}} = S\,\mathbb{1}_s + \Lambda \operatorname{tr}\left(\underline{\underline{e}}\right) + 2M\underline{\underline{e}} \tag{34}$$

To reduce the number of parameters, we choose:

- one membrane entropic "stiffness" k,
- a prestretch ξ.

which define completely the elastic parameters as follows

$$\mathcal{E}_{\mathrm{i}}(\xi) = k\sqrt{3}\left(1 - 1/\xi\right)^2 \tag{35a}$$

$$S(\xi) = k\sqrt{3}\left(1 - 1/\xi\right) \tag{35b}$$

$$\Lambda(\xi) = M(\xi) = \frac{\sqrt{3}}{4}k/\xi \tag{35c}$$

To express the weak-form of the equation, we simply need one scalar displacement variable u_θ and the following formula

$$\underline{\underline{e}} = \frac{1}{R}\begin{pmatrix} u_{\theta\theta} & 0 \\ 0 & u_{\varphi\varphi}\end{pmatrix} + \frac{1}{2R^2}\begin{pmatrix} u_{r\theta}^2 + u_{\theta\theta}^2 & 0 \\ 0 & u_{\varphi\varphi}^2\end{pmatrix} \tag{36}$$

$$\mathrm{tr}\left(\underline{\underline{e}}\right) = \frac{1}{R}\left[\left(u_{\theta\theta} + u_{\varphi\varphi}\right) + \frac{1}{2R}\left(u_{\theta\theta}^2 + u_{\varphi\varphi}^2 + u_{r\theta}^2\right)\right] \tag{37}$$

with

$$u_{r\theta} = -u_\theta \tag{38}$$

$$u_{\theta\theta} = \frac{\partial u_\theta}{\partial \theta} \tag{39}$$

$$u_{\varphi\varphi} = u_\theta \cot\theta \tag{40}$$

2 Equations proposal

I think what follows is not relevant for now

Tension balance

$$\boldsymbol{\nabla}_s \cdot \underline{\underline{N}} = \chi \boldsymbol{v} - \Delta P\, \boldsymbol{n} \tag{41}$$

$$\underline{\underline{N}} = 2\mu_s \underline{\underline{\epsilon}} + (\lambda_s - \mu_s)\left(\boldsymbol{\nabla}_s \cdot \boldsymbol{v}\right)\underline{\underline{P}} + e\sigma_a \underline{\underline{P}} \tag{42}$$

We'll choose $\lambda_s = 3\mu_s$ and $\mu_s = e\mu$ (Turlier 2014, Borja da Rocha 2022)

$$\underline{\underline{N}} = 2\mu\underline{\underline{\epsilon}} + 2e\mu\left(\boldsymbol{\nabla}_s \cdot \boldsymbol{v}\right)\underline{\underline{P}} + e\sigma_a \underline{\underline{P}} \tag{43}$$

and

$$\underline{\underline{\epsilon}} = \frac{1}{2}\left(\boldsymbol{\nabla}_s \boldsymbol{v} + \left(\boldsymbol{\nabla}_s \boldsymbol{v}\right)^T\right) \tag{44}$$

We will introduce Pressure as a Lagrange multiplier for volume

$$\Delta P \int \pi r^2(z)\, dz - V_0 \tag{45}$$

Thus we solve,

$$\boldsymbol{\nabla}_s \cdot \left[2e\mu\frac{1}{2}\left(\boldsymbol{\nabla}_s \boldsymbol{v} + \left(\boldsymbol{\nabla}_s \boldsymbol{v}\right)^T\right) + (2e\,\mu)\left(\boldsymbol{\nabla}_s \cdot \boldsymbol{v}\right)\underline{\underline{P}} + e\sigma_a \underline{\underline{P}}\right] = \chi \boldsymbol{v} \tag{46}$$

$$\begin{aligned} &\mu\Big(\left(\boldsymbol{\nabla}_s e\right)\left(\boldsymbol{\nabla}_s \boldsymbol{v} + \left(\boldsymbol{\nabla}_s \boldsymbol{v}\right)^T\right) + e\boldsymbol{\Delta}_s \boldsymbol{v}\Big) + \\ &2\mu\Big(\boldsymbol{\nabla}_s(e\,\boldsymbol{\nabla}_s \cdot \boldsymbol{v}) \cdot \underline{\underline{P}} + e\left(\boldsymbol{\nabla}_s \cdot \boldsymbol{v}\right)\boldsymbol{\nabla}_s \cdot \underline{\underline{P}}\Big) + \\ &\boldsymbol{\nabla}_s(\sigma_a\, e) \cdot \underline{\underline{P}} + \sigma_a\, e \boldsymbol{\nabla}_s \cdot (\underline{\underline{P}}) = \chi \boldsymbol{v} \end{aligned} \tag{47}$$

Usually the term ∇e would be zero, as well for $\nabla \cdot v$ in normal fluid mechanics. This will not be the case anymore. When e becomes a variable we will have a new additional term remaining, we have to take into account too the projector operator

$$\boldsymbol{\nabla}_s \cdot \underline{\underline{\boldsymbol{P}}} = -H\boldsymbol{n} \tag{48}$$

to get

$$\begin{aligned}
&\mu\Big((\boldsymbol{\nabla}_s e)\left(\boldsymbol{\nabla}_s \boldsymbol{v} + (\boldsymbol{\nabla}_s \boldsymbol{v})^T\right) + e\boldsymbol{\Delta}_s \boldsymbol{v}\Big) + \\
&2\mu\Big(\boldsymbol{\nabla}_s(e\,\boldsymbol{\nabla}_s \cdot \boldsymbol{v}) \cdot \underline{\underline{P}} - e\,(\boldsymbol{\nabla}_s \cdot \boldsymbol{v})\,H\boldsymbol{n}\Big) + \\
&\boldsymbol{\nabla}_s(\sigma_a\,e) \cdot \underline{\underline{P}} - \sigma_a\,e\,H\boldsymbol{n} = \chi\boldsymbol{v}
\end{aligned} \tag{49}$$

I guess that for a non-deforming surface all the normal contributions will have to be zero, otherwise it would deform. Now lets try to take out the normal contributions to see an approximation of our strong form in a rigid case

$$\mu\Big((\boldsymbol{\nabla}_s e)\left(\boldsymbol{\nabla}_s \boldsymbol{v} + (\boldsymbol{\nabla}_s \boldsymbol{v})^T\right) + e\boldsymbol{\Delta}_s \boldsymbol{v} + 2\boldsymbol{\nabla}_s(e\,\boldsymbol{\nabla}_s \cdot \boldsymbol{v}) \cdot \underline{\underline{P}}\Big) + \boldsymbol{\nabla}_s(\sigma_a\,e) \cdot \underline{\underline{P}} = \chi\boldsymbol{v} \tag{50}$$

The fact that we have a non constant e adds so many new terms,

$$\begin{aligned}
&\mu\Big((\boldsymbol{\nabla}_s e)\left(\boldsymbol{\nabla}_s \boldsymbol{v} + (\boldsymbol{\nabla}_s \boldsymbol{v})^T\right) + e\boldsymbol{\Delta}_s \boldsymbol{v} + 2(\boldsymbol{\nabla}_s e \boldsymbol{\nabla}_s \cdot \boldsymbol{v} + e\boldsymbol{\nabla}_s(\boldsymbol{\nabla}_s \cdot \boldsymbol{v})) \cdot \underline{\underline{P}}\Big) + \\
&(e\boldsymbol{\nabla}_s \sigma_a + \sigma_a \boldsymbol{\nabla}_s e) \cdot \underline{\underline{P}} = \chi\boldsymbol{v}
\end{aligned} \tag{51}$$

Honestly the flow being incompressible also is a non-negligible change, it might have even a bigger impact than e. Without the compressibility or e we recover our well-known equation for the in plane velocity (no normal velocity)

$$\mu e\boldsymbol{\Delta}_s \boldsymbol{v} + (e\boldsymbol{\nabla}_s \sigma_a) \cdot \underline{\underline{P}} = \chi\boldsymbol{v}. \tag{52}$$

Actin Turnover

$$\frac{\partial e}{\partial t} + \boldsymbol{\nabla}_s \cdot (e\boldsymbol{v}) = \frac{\partial e}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}_s e + e\boldsymbol{\nabla}_s \cdot \boldsymbol{v} = D_e \boldsymbol{\nabla}_s^2 e + k_d(e_0 - e) \tag{53}$$

thus

$$\frac{\partial e}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}_s e + e\boldsymbol{\nabla}_s \cdot \boldsymbol{v} = D_e \boldsymbol{\nabla}_s^2 e + k_d(e_0 - e). \tag{54}$$

$\boldsymbol{v}$ being the velocity of the surface, including surface and normal components $\boldsymbol{v} = \boldsymbol{v}_s + \boldsymbol{n} \cdot \boldsymbol{v} = \boldsymbol{v}_s + v_n$ k_d is 40s, $e_0 = 0.01\mu m$

Rac activity

$$\frac{\partial R}{\partial t} + \boldsymbol{\nabla}_s \cdot (R\boldsymbol{v}) = \frac{\partial R}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}_s R + R\boldsymbol{\nabla}_s \cdot \boldsymbol{v} = D_R \boldsymbol{\nabla}_s^2 R + \alpha_0 \frac{k_R^2}{k_R^2 + \rho^2} - d_R(R - R_0) \tag{55}$$

Rho activity

$$\frac{\partial \rho}{\partial t}+\boldsymbol{\nabla}_s\cdot(\rho\boldsymbol{v}) = \frac{\partial \rho}{\partial t}+\boldsymbol{v}\cdot\boldsymbol{\nabla}_s\rho+\rho\boldsymbol{\nabla}_s\cdot\boldsymbol{v} = D_\rho\boldsymbol{\nabla}_s^2\rho+\beta_0\frac{k_\rho^2}{k_\rho^2+R^2}-d_\rho(\rho-\rho_0) \tag{56}$$

Mechanochemical couplings

$$\chi = \chi_0 + (\delta\chi)R \tag{57}$$

$$\sigma_a = \max\Big(\sigma_a^0 + \sigma_\rho^0\,\rho - \sigma_R^0\,R\ ,\ 0\Big) \tag{58}$$

$$\tag{59}$$

Hydrodynamic length

$$\lambda_H = \sqrt{\frac{2\mu e}{\chi}} \tag{60}$$

$$\frac{\partial R}{\partial t} = \Big(\alpha_0 + \alpha\,\mathrm{th}(\rho_b - \rho_{\mathrm{th}})\Big)\frac{k_R^2}{k_R^2+\rho^2} - d_R R + D_R\nabla^2 R, \tag{61a}$$

$$\frac{\partial \rho}{\partial t} = \Big(\beta_0 + \beta\,\mathrm{th}(\sigma - \sigma_{th})\Big)\frac{k_\rho^2}{k_\rho^2+R^2} - d_\rho\rho + D_\rho\nabla^2\rho, \tag{61b}$$

$$\eta\nabla^2 v + \sigma_a^0\nabla\left(\frac{\rho}{k_R}\right) - \chi\,\rho_b\,v = 0, \tag{61c}$$

No incompressibility in the surface divergence, update in equations this

$$\frac{\partial R}{\partial t} + \boldsymbol{\nabla}_s\cdot(R\boldsymbol{v}) = \frac{\partial R}{\partial t} + \boldsymbol{v}\cdot\boldsymbol{\nabla}_s R + R\boldsymbol{\nabla}_s\cdot\boldsymbol{v} \tag{62}$$

2.1 Weak form

The weak form for the rigid surface equation.

Introducing the test function w. Laplacian operator becomes the product of two gradients using the integration by parts and the fact that the test function w is taken so that it is zero at the BCs.

$$\int\Big[\mu\Big(w(\boldsymbol{\nabla}_s e)(\boldsymbol{\nabla}_s^{Sym}\boldsymbol{v}) + e\boldsymbol{\nabla}_s\boldsymbol{v}\boldsymbol{\nabla}_s w + 2(w\boldsymbol{\nabla}_s e\boldsymbol{\nabla}_s\cdot\boldsymbol{v} + w\,e\boldsymbol{\nabla}_s(\boldsymbol{\nabla}_s\cdot\boldsymbol{v}))\cdot\underline{\underline{P}}\Big)+ \tag{63}$$

$$w(e\boldsymbol{\nabla}_s\sigma_a + \sigma_a\boldsymbol{\nabla}_s e)\cdot\underline{\underline{P}}\Big]dS = \int\Big[w\,\chi\,\boldsymbol{v}\Big]dS \tag{64}$$

in red the part I should check again for possible improvements.

3 Introduction

4 Equations

$$\frac{\partial R}{\partial t} = \Big(\alpha_0 + \alpha\,\mathrm{th}(\rho_b - \rho_{\mathrm{th}})\Big)\frac{k_R^2}{k_R^2+\rho^2} - d_R R + D_R\nabla^2 R, \tag{65a}$$

$$\frac{\partial \rho}{\partial t} = \Big(\beta_0 + \beta\,\mathrm{th}(\sigma - \sigma_{th})\Big)\frac{k_\rho^2}{k_\rho^2+R^2} - d_\rho \rho + D_\rho\nabla^2 \rho, \tag{65b}$$

$$\sigma_0\nabla^2 x - \chi\,\rho_b\Big(\dot{x} - v\Big) = 0, \tag{65c}$$

$$\eta\nabla^2 v + \sigma_a^0\nabla\left(\frac{\rho}{k_R}\right) - \chi\,\rho_b\Big(v - \dot{x}\Big) = 0, \tag{65d}$$

$$\frac{\partial \rho_b}{\partial t} + \nabla(v\cdot\rho_b) = k_{\mathrm{on}}\rho_u - k_{\mathrm{off}}\rho_b, \tag{65e}$$

$$\frac{\partial \rho_u}{\partial t} + \nabla\big(\dot{x}\cdot\rho_u\big) = -k_{\mathrm{on}}\rho_u + k_{\mathrm{off}}\rho_b + D\nabla^2\rho_u, \tag{65f}$$

4.1 Boundary Conditions

$$\nabla R\Big|_0^L = 0 \tag{66a}$$

$$\nabla \rho\Big|_0^L = 0 \tag{66b}$$

$$x\Big|_0 = x_{protrusion} \quad \& \quad x\Big|^L = 0 \tag{66c}$$

$$v\Big|_0 = v_{protrusion} \quad \& \quad v\Big|^L = 0 \tag{66d}$$

$$\nabla \rho_b\Big|_0^L = 0 \tag{66e}$$

$$\nabla \rho_u\Big|_0^L = 0 \tag{66f}$$

4.2 Weak Formulation

$$\int_0^L w(\xi,t)\left[\sigma_0\frac{\partial^2}{\partial\xi^2}x(\xi,t) - \frac{1}{\lambda^2}\rho_b(\xi,t)\big(\dot{x}(\xi,t) - v(\xi,t)\big)\right]d\xi = 0. \tag{67}$$

Integrating by parts we obtain

$$\begin{aligned}\int w\sigma_0\frac{\partial}{\partial\xi}\frac{\partial x}{\partial\xi}d\xi \quad &= \textstyle\int k\frac{\partial}{\partial\xi}\Big(w\frac{\partial\, x}{\partial\xi}\Big)\partial\xi - \int\frac{1}{\lambda^2}\frac{\partial x}{\partial\xi}\cdot\frac{\partial w}{\partial\xi}d\xi \\ &= \textstyle\sigma_0\Big(w\frac{\partial x}{\partial\xi}\Big)\Big|_0^L - \int k\frac{\partial x}{\partial\xi}\cdot\frac{\partial w}{\partial\xi}d\xi\end{aligned} \tag{68}$$

where the constant term valued at the boundaries will be zero from the test function w being zero at the boundaries. Using this integration by parts, we write

$$\int_0^L \left[-k \frac{\partial x}{\partial \xi} \cdot \frac{\partial w}{\partial \xi} - \frac{w}{\lambda^2} \rho_b(\xi,t)\big(\dot{x}(\xi,t) - v(\xi,t)\big) \right] d\xi = 0. \tag{69}$$

We do this and other for all the equations, leading to the following set of weak equations

$$\int w \frac{\partial R}{\partial t} d\xi = \int \Big[\Big(\alpha_0 + \alpha \, \mathrm{th}(\rho_b - \rho_{\mathrm{th}}) \Big) \frac{k_R^2}{k_R^2 + \rho^2} - d_R R + D_R \nabla w \nabla R \Big] d\xi, \tag{70a}$$

$$\int w \frac{\partial \rho}{\partial t} \, d\xi = \int \Big[w \Big(\beta_0 + \beta \, \mathrm{th}(\sigma - \sigma_{th}) \Big) \frac{k_\rho^2}{k_\rho^2 + R^2} - w \, d_\rho \rho + D_\rho \nabla w \nabla \rho \Big] d\xi, \tag{70b}$$

$$\int_0^L \left[-k \frac{\partial x}{\partial \xi} \cdot \frac{\partial w}{\partial \xi} - \frac{w}{\lambda^2} \rho_b \big(\dot{x} - v \big) \right] d\xi = 0, \tag{70c}$$

$$\int_0^L \Big[-\eta \nabla w \nabla v + w \frac{\partial \sigma_a}{\partial \xi} - w \, \chi \, \rho_b(\xi) \Big(v - \dot{x} \Big) \Big] d\xi = 0, \tag{70d}$$

$$\int \Big[w \frac{\partial \rho_b}{\partial t} + w \nabla (v \cdot \rho_b) \Big] d\xi = \int w \Big[k_{\mathrm{on}} \rho_u - k_{\mathrm{off}} \rho_b \Big] d\xi, \tag{70e}$$

$$\int \Big[w \frac{\partial \rho_u}{\partial t} + w \nabla \big(\dot{x} \cdot \rho_u \big) \Big] d\xi = \int w \Big[- k_{\mathrm{on}} \rho_u + k_{\mathrm{off}} \rho_b + D \nabla w \nabla \rho_u, \Big] d\xi \tag{70f}$$

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